

JEE MAIN REVISION MASTER: UNITS & MEASUREMENT AND DIMENSIONS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Dimensional Analysis & Principle of Homogeneity	7 PYQs	Dimensions
Moderate	Error Analysis & Significant Figures	3 PYQs	Errors

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Dimensional Analysis & Principle of Homogeneity	$[G] = M^{-1}L^3T^{-2}$, $[h] = ML^2T^{-1}$, $[\mu_0] = MLT^{-2}A^{-2}$
Error Analysis & Significant Figures	$\frac{\Delta Z}{Z} = a\frac{\Delta A}{A} + b\frac{\Delta B}{B} + c\frac{\Delta C}{C}$ $\text{for } Z = \frac{A^a B^b}{C^c}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Dimensional Analysis & Principle of Homogeneity (Moderate Yield — 7 PYQs)

Core Principle: Application of dimensions to check equation consistency and derive relations between physical quantities.

■ **Caution / JEE Pitfall:** Trigonometric, logarithmic, and exponential arguments must strictly be dimensionless.

■ **Speed Trick:** Equate powers of M, L, T on both sides to find unknown exponents.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match the physical quantities given in List-I with dimensions expressed in terms of mass (M), length (L), time (T) and electric current (A) ..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Which of the following expression can be deduced on the basis of dimensional analysis? (All symbols have their usual meanings)..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Dimensional formula for activity of a radioactive substance is..." [Ans: C]

3.2 Error Analysis & Significant Figures (Moderate Yield — 3 PYQs)

Core Principle: Maximum fractional and percentage errors in combination of quantities and screw gauge / vernier calipers reading.

■ **Caution / JEE Pitfall:** Errors always ADD up in worst-case analysis; never subtract fractional errors.

■ **Speed Trick:** Least count = Pitch / (Number of circular divisions). Total reading = MSR + (CSR × LC) - (Zero Error).

Representative Source PYQs:

- **[2022 Shift] Q#1 (MCQ):** "The Vernier scale of a travelling microscope has 50 divisions which coincides with 49 main scale divisions. If each main scale division is \$...\$ " **[Ans: N/A]**
- **[2020 Shift] Q#2 (MCQ):** "A cylindrical wire has a mass $(0.3 \pm 0.003)\text{g}$, radius $(0.5 \pm 0.005)\text{mm}$ and length $(6 \pm 0.06)\text{cm}$. The maximum percentage error in ..." **[Ans: N/A]**
- **[2017 Shift] Q#1 (MCQ):** "A substance of mass 49.53 g occupies 1.5cm^3 of volume. The density of the substance (in gcm^{-3}) with correct number of significant..." **[Ans: D]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match the physical quantities given in List-I with dimensions expressed in terms of mass (M), length (L), time (T) and electric current (A) given in List-II. List-I List-II (a) Torque (i) $[M^{-1}L^{-2}T^4A^2]$ (b) Gravitational constant (ii) $[M^1L^2T^{-1}]$ (c) Capacitance (iii) $[M^{-1}L^3T^{-2}]$ (d) Planck's constant (iv) $[M^1L^2T^{-2}]$ Codes:

- (A) a - iv, b - ii, c - iii, d - i
- (B) a - iv, b - iii, c - i, d - ii
- (C) a - iv, b - i, c - iii, d - ii
- (D) a - ii, b - i, c - iii, d - iv

Correct Answer: None

Solution Analysis: For each quantity, we write its dimensional formula. Torque is given by Torque = Force × distance. Force has dimensions $[MLT^{-2}]$. So, $[\text{Torque}] = [MLT^{-2}][L] = [ML^2T^{-2}]$. So, torque matches (iv). For gravitational constant G, from Newton's law of gravitation, $F = \frac{Gm_1m_2}{r^2}$. So, $G = \frac{Fr^2}{m_1m_2}$. Now, $[G] = \frac{[MLT^{-2}][L^2]}{[M^2]}$. $[G] = [M^{-1}L^3T^{-2}]$. So, gravitational constant matches (iii). Capacitance is $C = (Q / V)$. Charge has dimensions $[Q] = [AT]$. Potential difference is $V = \frac{\text{Work}}{\text{Charge}}$.

Benchmark Problem 2: KCET 2022 (Q#1)

The Vernier scale of a travelling microscope has 50 divisions which coincides with 49 main scale divisions. If each main scale division is 0.5mm , then the least count of the microscope is

- (A) 0.5mm
- (B) 0.01mm
- (C) 0.5cm
- (D) 0.01cm

Correct Answer: None

Solution Analysis: The least count of a measuring instrument is the smallest value that can be measured with the instrument and is calculated by the formula: $\text{Least Count} = (\text{Value of one main scale division} / \text{Number of divisions on the Vernier scale})$ In the given travelling microscope, there are 50 divisions on the Vernier scale which coincides with 49 main scale divisions. Each main scale division is of 0.5mm . First, we need to find out the value of one main scale division: Given that one main scale division is 0.5mm , we keep this value for the calculation. Next, to calculate the l